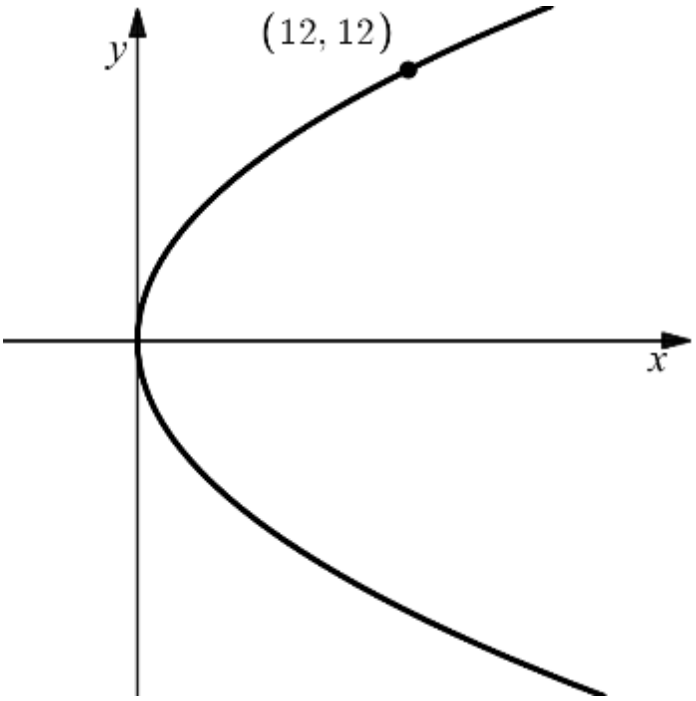
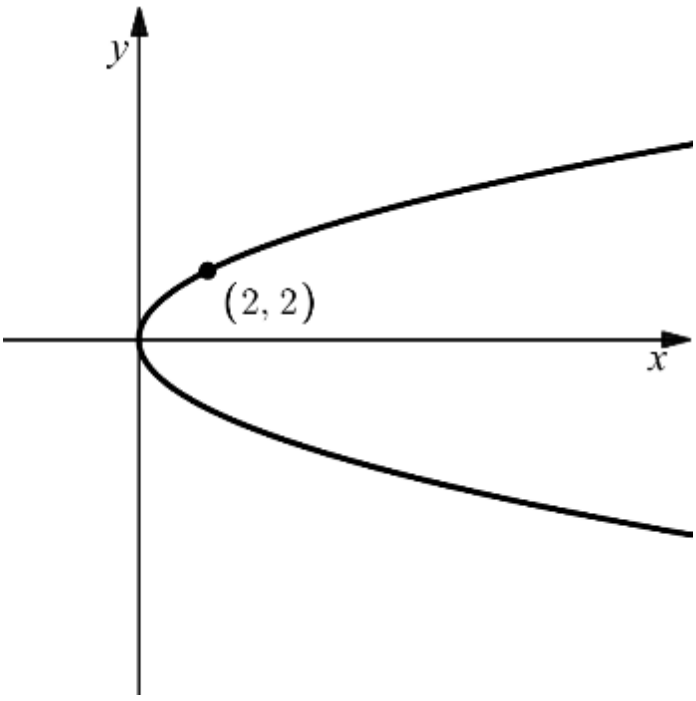
1.21 Graph Sketching: Parabolas

Worked Solutions

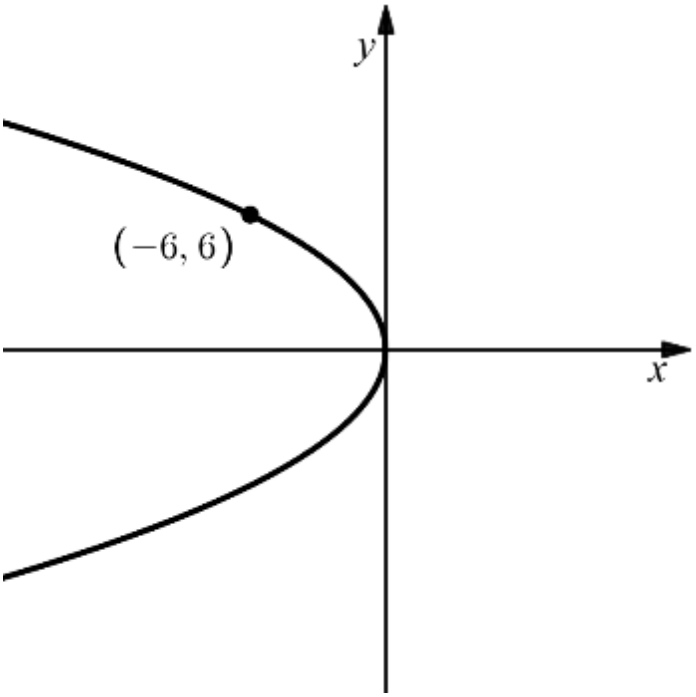
1 a)



1 b)



1 c)



2 a)

$(\sqrt{12})^2 = 3a$

$$\therefore a = 4$$

2 b)

$(-\sqrt{2})^2 = -a$

$$\therefore a = -2$$

2 c)

$6^2 = 6\sqrt{6}x$

$$\therefore a = \sqrt{6}$$

3 a)

intersection of $x = \frac{1}{4}y^2$ with $y = x$:

$$y = \frac{1}{4}y^2$$
$$\Rightarrow y(1 - \tfrac{1}{4}y) = 0$$
$$\therefore y = 0 \text{ or } y = 4$$

$y^2 = 4x$ and $P(4, 4)$

3 b)

intersection of $x = 3y^2$ with $y = x$:

$$\begin{aligned} y &= 3y^2 \\ \Rightarrow y(1-3y) &= 0 \\ \therefore y = 0 \text{ or } y &= \frac{1}{3} \end{aligned}$$

$y^2 = 4x$ and $P(4, 4)$

$y^2 = \frac{1}{3}x$ and $P(\frac{1}{3}, \frac{1}{3})$

3 c)

intersection of $x = ay^2$ with $y = x$:

$$\begin{aligned} y &= ay^2 \\ \Rightarrow y(1-ay) &= 0 \\ \therefore y = 0 \text{ or } y &= \frac{1}{a} \end{aligned}$$

$y^2 = 4x$ and $P(4, 4)$

$y^2 = \frac{1}{a}x$ and $P(\frac{1}{a}, \frac{1}{a})$

4 a)

$(2, 4)$

4 b)

$(4, 2)$

4 c)

$(5, -1)$

4 d)

$(-1, -5)$

5 a)

$a = 3$ and $b = -4$

5 b)

$(y-3)^2 = x + 4$

When $y = 0$, $x = 5$

When $x = 0$, $y = 3 \pm 2$ so $y = 1$ or $y = 5$

$P(0, 5)$, $Q(0, 1)$ and $R(5, 0)$